MIL-D-45495A (MU)
5 March 1969
SUPERSEDING
MIL-D-45495 (ORD)
24 March 1960 and
MIL-D-45494A (ORD)
28 February 1962

MILITARY SPECIFICATION

DETONATOR, ELECTRIC, M84
PARTS FOR AND LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the inspection of the parts and the loading, assembling and packing of one type of electric initiating device designated as Detonator, Electric, M84 (T77).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-650 - Explosives; Sampling, Inspection and Testing
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1330

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:13Z
Check the source to verify that this is the current version before use.

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

8798698 - Box, Set-Up, Packing, Ammunition, for Detonators, Electric
8798699 - Box, Packing, Ammunition for Detonators, Electric
8800784 - Ammunition Container, Metal for Detonators, Electric
8838107 - Detonator, Electric, M84 Assembly

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

ET - 8838107 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.- Material and parts shall be in accordance with applicable specifications and drawings.

3.2 Assembly.- The detonator assemblies shall comply with all requirements specified on Drawing (dwg.) 8838107 and with all requirements specified in applicable specifications.

3.3 Moisture content

3.3.1 Lead azide.- The moisture content of the lead azide at the loading station, at the time of loading the detonators, shall not exceed 0.50 percent, when tested as specified in 4.4.1.

3.3.2 Pentaerythrite tetranitrate (PETN).- The moisture content of the PETN at the loading station, at the time of loading the detonators, shall not exceed 0.50 percent when tested as specified in 4.4.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:13Z
Check the source to verify that this is the current version before use.

3.4 Porosity.- The plug assembly (dwg. 8860881) shall show no evidence of porosity as indicated by air bubbles when tested as specified in 4.4.2.

3.5 Functioning.- The detonators shall comply with the following:

3.5.1 Time.- The detonator shall fire on the first attempt within 100 microseconds when tested as specified in 4.4.3.1.

3.5.2 Output.- The detonator shall fire on the first attempt and shall cause perforation of the lead disc to a diameter of not less than 5/16 inch, minimum (min.), when tested as specified in 4.4.3.2.

3.5.3 Failure to fire (Duds).- The detonator shall fire on the first attempt when tested as specified in 4.4.3.3.

3.6 Waterproofness.- The detonator shall comply with the requirements of 3.5.2 and 3.5.3 when tested as specified in 4.4.3.2.

3.7 Workmanship.- All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. All packing components shall be dry.

3.8 First article inspection.- This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:13Z
Check the source to verify that this is the current version before use.

supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.- At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review, except that a statement of findings for the moisture content of the lead azide and PETN shall be submitted with each lot submitted for acceptance.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.- The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of twenty-five (25) plug assemblies and 25 each of

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:13Z
Check the source to verify that this is the current version before use.

every component part which has been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and he shall submit a record of this inspection with the sample, including certificates of conformance for materials. A sample containing known defects shall not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.- The plug assemblies and component parts will be subjected by the Government to any or all of the examinations specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.- If any plug assembly or component part fails to comply with any or all of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an item to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation.- The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:13Z
Check the source to verify that this is the current version before use.

specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded detonators shall contain:

a. Metal parts from not more than one lot interfix number from one supplier.

b. PETN from not more than one lot from one manufacturer.

c. Lead azide of the same type from not more than two (2) lots produced by the same manufacturer in accordance with the same specification and same specification revision and which have subsequently been subjected to uniformly controlled drying and operating conditions at the loading plant.

4.3.2 Examination.- The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:13Z
Check the source to verify that this is the current version before use.

4.3.2.1 Cup (see dwg. 8838115 covering a detail of dwg. 8838107).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:			
1.	Crack, split or cut.....	Visual	01001
Major:	AQL 1.00 percent		
101.	Thickness through bottom.....	Gage	01002
102.	Inside diameter, maximum (max.).	Gage	01003
103.	Outside diameter, min.....	Gage	01004
104.	Total length.....	Gage	01005
105.	Dent, deep scratch, groove or distortion.....	Visual	01006
Minor:	AQL 1.00 percent		
201.	Finish improper.....	Visual	01007
202.	Evidence of poor workmanship other than critical above (see 3.7).....	Visual	01008

4.3.2.2 Ferrule (see dwg. 8838110 covering a detail of dwg. 8838107).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Inside diameter, min.....	Gage	02001
102.	Outside diameter, max.....	Gage	02002
Minor:	AQL 1.50 percent		
201.	Length.....	Gage	02003
202.	Chamfer missing.....	Visual	02004
203.	Evidence of poor workmanship (see 3.7).....	Visual	02005

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:13Z
Check the source to verify that this is the current version before use.

4.3.2.3 Plug (see dwg. 8860883 covering a detail of dwg. 8838107).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Depth of hole counterbore.....	Gage	03001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7).....	Visual	03002

4.3.2.4 Plug assembly (see dwg. 8860881 covering a detail of dwg. 8838107).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Diameter of pin.....	Gage	04001
102.	Length of pin protrusion.....	Gage	04002
103.	Concentricity of pin with plug diameter.....	Gage	04003
104.	Protective finish missing.....	Visual	04004
105.	Evidence of poor workmanship on face of plug (see 3.7).....	Visual	04005
Minor:	AQL 1.50 percent		
201.	Large diameter of plug, min.....	Gage	04006
202.	Length of plug.....	Gage	04007
203.	Evidence of poor workmanship other than on face of plug (see 3.7).....	Visual	04008

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:13Z
Check the source to verify that this is the current version before use.

4.3.2.5 Plug (see dwg. 8838109 covering a detail of dwg. 8838107).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Hole surface not coated.....	Visual	05001
Minor:	AQL 1.00 percent		
201.	Outside diameter, min.....	Gage	05002
202.	Evidence of poor workmanship (see 3.7).....	Visual	05003

4.3.2.6 Plug assembly (see dwg. 8838111 covering a detail of dwg. 8838107).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Length of pin protrusion.....	Gage	06001
102.	Diameter of pin.....	Gage	06002
103.	Excess bonding material in gap..	Visual	06003
104.	Evidence of poor workmanship on face of plug (see 3.7).....	Visual	06004
Minor:	AQL 1.50 percent		
201.	Large diameter of plug, min.....	Gage	06005
202.	Length of plug.....	Gage	06006
203.	Evidence of poor workmanship other than on face of plug (see 3.7).....	Visual	06007

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:13Z
Check the source to verify that this is the current version before use.

4.3.2.7 Bridge and plug assembly (see dwg. 8838117 covering a detail of dwg. 8838107).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Spot charge missing	Visual	07001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	07002

4.3.2.8 Assembly (see dwg. 8838107).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Crack, split or cut through cup	Visual	08001
Major:	AQL 1.00 percent		
101.	Total length, max	Gage	08002
102.	Diameter	Gage	08003
103.	Contact surface of pin not clean	Visual	08004
104.	Crimp not full 360 degrees	Visual	08005
105.	Joint seal missing or exceeding maximum limit	Visual	08006
Minor:	AQL 1.50 percent		
201.	Length to end of cup	Gage	08007
202.	Cup contains deep scratches or grooves	Visual	08008
203.	Evidence of poor workmanship other than critical above (see 3.7)	Visual	08009

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:13Z
Check the source to verify that this is the current version before use.

4.3.2.9 Sealed metal container (see dwg. 8800784).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Contents exposed.....	Visual	09001
102.	Sealing strip incomplete or badly wrinkled.....	Visual	09002
Minor:	AQL 1.50 percent		
201.	Contents move when shaken.....	Manual	09003
202.	Marking missing, incorrect or illegible.....	Visual	09004
203.	Bare spot on exterior other than slight scratches.....	Visual	09005

4.3.2.10 Box, Set-Up, Packing (see dwg. 8798698).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Waterproof seal broken by puncture, cut or tear.....	Visual	10001
102.	Sealing strip incomplete or badly wrinkled.....	Visual	10002
Minor:	AQL 1.00 percent		
201.	Contents move when shaken.....	Manual	10003
202.	Marking missing, incorrect or illegible.....	Visual	10004

4.3.2.11 Sealed wood packing box (see dwg. 8798699).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:13Z
Check the source to verify that this is the current version before use.

Major:	AQL 0.65 percent		
101.	Strapping missing, broken or loose.....	Visual-Manual	11001
102.	Contents exposed.....	Visual	11002
103.	Board broken or split.....	Visual	11003
104.	DOD symbol missing or illegible.	Visual	11004

Minor: None defined

4.3.3 Testing.- Unless otherwise specified herein, the following tests shall be performed using MIL-STD-105 for sampling plan requirements.

4.3.3.1 Moisture content (see 3.3) - Major defect

TABLE I

Explosive	Code No.
Lead azide	12001
PETN	12002

The contractor shall provide adequate controls to insure that the explosives comply with the requirements. For verification, the contractor shall select a sample for test from that quantity of each explosive used in each eight hours of loading production. A composite sample shall not be used. If the moisture content of a sample exceeds the requirements and loading has not begun, that quantity of explosive represented by the sample shall be rejected. If components have been loaded with explosive containing excessive moisture, the remaining unloaded explosive and the loaded components shall be rejected. The test procedure shall be as specified in 4.4.1.

4.3.3.2 Porosity (see 3.4) - Major defect - Code No. 04009.- The sampling plan shall be in accordance with Code Letter H and with an AQL of 0.40 percent. The test shall be performed as specified in 4.4.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:13Z
Check the source to verify that this is the current version before use.

4.3.3.3 Functioning and waterproofness

4.3.3.3.1 Sampling.- One hundred and eighty-five (185) detonator assemblies shall be selected from each lot for test. The sequence of testing and the number of samples to be subjected to each test shall be as specified in Table II.

TABLE II

Sequence	Test	Classification	Code No.	No. of Samples	Test Para.
1.	Time (see 3.5.1)	Major	08011	35	4.4.3.1
2.	Output (see 3.5.2) and waterproofness (see 3.6)	Major	08012	50	4.4.3.2
3.	Duds (see 3.5.3)	Major	08013	100*	4.4.3.3

*Duds that occur in the time or output and waterproofness phase of testing shall be recorded and replaced from this sample.

4.3.3.3.2 Rejection.- The lot shall be rejected if any of the following test results occur:

a. Time.- The mean functioning time ($\bar{X}$) plus the product of 2.27 and the standard deviation of the sample (s) exceeds 100 microseconds (see 6.6).

b. Output and waterproofness.- One or more detonator assemblies fail to produce a perforation of the required size in the lead disc.

c. Duds.- The total number of detonator assemblies that fail to fire on the first attempt in the time, output and waterproofness, and dud tests equals or exceeds two (2).

4.3.3.4 Plug assembly push test (see dwgs. 8860881 and 8838111) - Major defect - Code Nos. 04010 and 06008.- The sampling plan shall be in accordance with Code Letter L with an AQL of 0.40 percent. The test shall be performed as specified in 4.4.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:13Z
Check the source to verify that this is the current version before use.

4.3.3.5 Resistance between pin and plug (see dwgs. 8860881 and 8838111) - Major defect - Code Nos. 04011 and 06009.- The test shall be performed 100 percent in accordance with the procedure specified in 4.4.5. Any assembly that fails to comply with the requirement shall be classed defective and removed from the lot.

4.3.3.6 Resistance between pin and cup (see dwg. 8838107) - Major defect - Code No. 08014.- The test shall be performed 100 percent in accordance with the procedure specified in 4.4.6. Any assembly that fails to comply with the requirements shall be classed defective and removed from the lot.

4.3.4 Inspection equipment.- Equipment Tabulation Number ET - 8838107 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.- Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures

4.4.1 Moisture content of lead azide and PETN.- The moisture content of the lead azide and PETN shall be determined in accordance with Method 101.5 of MIL-STD-650.

4.4.2 Porosity of plug assembly (see dwg. 8860881).- The plug assembly shall be immersed in an enclosure containing water and a suitable wetting agent. The enclosure shall be evacuated to an absolute pressure equal to 1.3 inches of mercury for one minute, or until air bubbles cease to be given off by the water, whichever is longer. The immersed plug assembly shall be observed for evidence of leakage, indicated by the bubbles emanating from between the pin and plug.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:13Z
Check the source to verify that this is the current version before use.

4.4.3 Functioning.- Unless otherwise specified herein, detonator assemblies used for the following tests shall be temperature conditioned to seventy (70) plus or minus five (5) degrees Fahrenheit (F.) for two (2) hours, prior to testing.

4.4.3.1 Time.- The detonator assemblies shall be assembled in the test fixture specified in 4.3.4 and fired by applying the electrical energy from a sixteen (16) microfarad capacitor max., charged to a maximum of 2.5 volts. The elapsed time between the application of the energy and detonator functioning shall be measured for acceptance determination. Detonators that fail to fire shall be recorded and the replacements tested.

4.4.3.2 Output and waterproofness.- The detonator assemblies shall be immersed in water at a two (2) to three (3) inch depth for forty-eight (48) hours, minimum. The water temperature shall be maintained at seventy (70) plus or minus ten (10) degrees F. for the immersion period. At the end of the immersion time, the detonator assemblies shall be removed from the water, assembled into the applicable test fixture and fired by applying the same energy specified in 4.4.3.1. The size of the lead disc perforation shall be measured to determine compliance with the requirement. Detonators that fail to fire shall be recorded and the replacements tested.

4.4.3.3 Duds.- The detonator assemblies shall be assembled in the test fixture specified in 4.3.4 and fired as specified in 4.4.3.1. Each detonator assembly shall be observed to determine compliance with the requirements.

4.4.4 Plug assembly push test.- The plug assembly shall be assembled in the test fixture specified in 4.3.4. The required axial force shall be applied to the pin in the direction indicated. The assembly shall be observed during the test to determine compliance with the requirement.

4.4.5 Resistance between pin and plug.- The plug assembly shall be connected in an electrical circuit containing a 325 minimum volt direct current (d.c.) power source and a resistance measuring instrument accurate within 2 percent for the full scale. The pin shall be used as one electrode and the plug as the other. The meter shall be observed for compliance with the requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:13Z
Check the source to verify that this is the current version before use.

4.4.6 Resistance between the pin and cup.- The detonator assemblies shall be connected to the test equipment specified in 4.3.4 and the electrical resistance measured. Not more than one milliampere of current shall be applied. This test shall be conducted within a suitable barricade.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C.- The detonators shall be preserved and packaged in accordance with dwg. 8800784.

5.1.2 Intermediate packaging.- The detonators shall be packaged in accordance with dwg. 8798698.

5.2 Packing

5.2.1 Level C - minimum military pack.- The containers shall be packed in accordance with dwg. 8798699.

5.3 Markings

5.3.1 Unit, intermediate and exterior containers shall be marked in accordance with dwgs. 8800784, 8798698 and 8798699.

6. NOTES

6.1 Ordering data.- Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards.- Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.- The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers

measuring instrument accurate within 2 percent for the full scale. The pin shall be used as one electrode and the plug as the other. The meter shall be observed for compliance with the requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:13Z
Check the source to verify that this is the current version before use.

Lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.5).

6.3 Inspection equipment.- The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).- Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.- The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.- Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:13Z
Check the source to verify that this is the current version before use.

equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.- Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.- All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND15. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:13Z
Check the source to verify that this is the current version before use.

6.4 Submission of ammunition data cards.- In addition to the normal distribution of records, one copy of all ammunition data cards shall be forwarded to each of the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey 07801
ATTN: (1) SMUPA-NDS
(2) SMUPA-DC.

6.5 Check test for possible deterioration.- The following test is designed to prevent detonators that have deteriorated being assembled into major ammunition items. The test shall be performed on detonators from each lot which have been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are assembled does not require that the check test be performed, the contracting officer should insert in the contract a provision for the performance of the check test. The contracting officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration. Two hundred (200) detonator assemblies shall be selected from each lot and tested as follows:

6.5.1 Functioning time and duds.- Forty (40) detonators shall be subjected to the test specified in 4.4.3.1. Observation shall be made for the functioning time and all duds that may occur shall be recorded. Duds will be replaced by detonators from the 200 selected. If the mean time ($\bar{X}$) plus 2.03 times the standard deviation (s) exceeds 100 microseconds, the lot shall be rejected. The standard deviation shall be computed from the formula in 6.6 or its equivalent. Duds shall not enter into the calculation for standard deviation.

6.5.2 Waterproofness, output, and duds.- Fifty (50) detonators shall be subjected to the test specified in 4.4.3.2. Observation shall be made of the perforation in the disc, and the duds that may occur shall be recorded. Duds shall be replaced by detonators from the 200 selected. If two (2) or more detonators fail to adequately perforate the lead disc, the lot shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:13Z
Check the source to verify that this is the current version before use.

6.5.3 Duds.- The remainder of the detonator assemblies selected in 6.5 shall be subjected to the test specified in 4.4.3.3. The lot shall be rejected if the total number of duds accumulated in 6.5.1, 6.5.2 and 6.5.3 equals or exceeds four (4).

6.6 Calculation of sample standard deviation.- The following formula or equivalent shall be used in calculating the sample standard deviation:

$$s = \sqrt{\frac{\Sigma(X - \overline{X})^2}{n-1}}$$

where:

s = sample standard deviation.
$\underline{x}$ = numerical value of individual observations.
$\overline{x}$ = average value of the sample.
n = sample size.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1330-A-204

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:13Z
Check the source to verify that this is the current version before use.